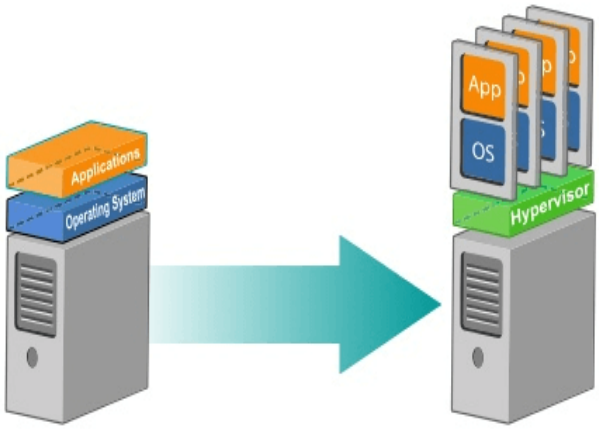


CLOUD COMPUTING COURSE

Lecture10-Memory and IO Virtualization

ABSTRACT

Lecture10 Introduces Memory and IO Virtualization.

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Memory Virtualization

Memory Virtualization

Memory virtualization decouples volatile random access memory (RAM) resources from individual systems in the data center and then aggregates those resources into a virtualized memory pool available to any computer in the cluster.

Why Memory Virtualization?

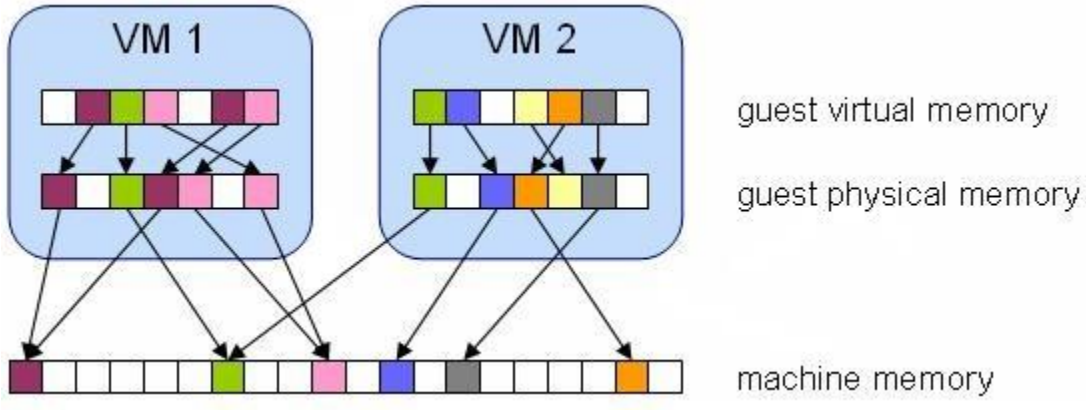
Applications can take advantage of a very large amount of memory to:

1. Improve overall performance.
2. Improve system utilization.
3. Increase memory usage efficiency.

Memory addresses in Virtualization

- **Guest virtual memory:** presented to applications by the guest OS.
- **Guest physical memory:** presented to the VM by Hypervisor.
- **Host machine memory:** provides a contiguous, zero-based, addressable memory space for use by the VM.

Address Binding is the process of mapping from one address space to another address space.



Address Binding:

1. In the virtual machine, the **guest operating system's** page tables maintain the mapping from **guest virtual memory (VA)** to **guest physical memory (PA)**.

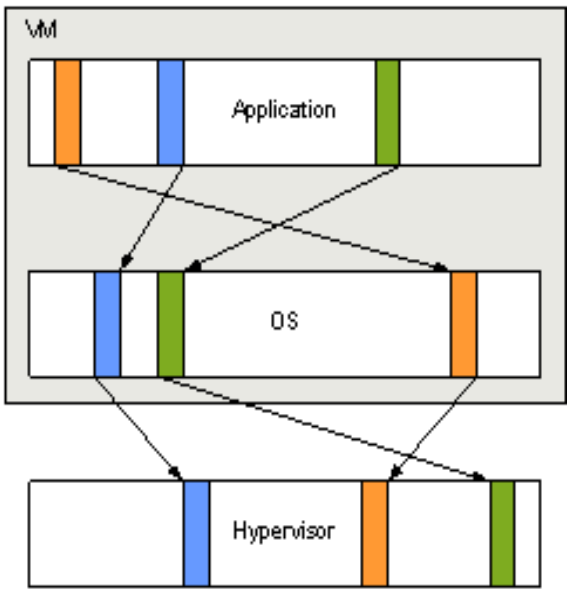
(VA→PA)

2. The **Hypervisor** then mapping from the **guest physical memory (PA)** to the physical machine memory **(MA)** on the underlying machine.

(PA→MA)

- Each virtual machine sees a contiguous, zero-based, addressable physical memory space.
- The underlying machine memory on the server used by each virtual machine is not necessarily contiguous.





I/O Virtualization

With compute virtualization, a large number of VMs can be created on a single host, and the VMs all need to access the I/O devices of this host. However, I/O devices are limited. I/O device sharing among multiple VMs requires VMM.

VMM intercepts access requests from VMs to I/O devices, simulates I/O devices using software, and responds to I/O requests.

This way, multiple VMs can access I/O resources concurrently.

I/O virtualization can be implemented in the following methods:

1. Full virtualization
2. Paravirtualization
3. Hardware-assisted virtualization. Hardware-assisted virtualization is the mainstream technology for I/O virtualization.

1. Full virtualization

The Mechanism

1. VMM virtualizes I/O devices for VMs. When a VM initiates an I/O request to an I/O device, VMM intercepts the request sent by the VM.
2. Then sends the real access request to the physical device for processing.

Full Virtualization Characteristics

- No matter which type of OS is used by the VM, the OS does not need to be modified for I/O virtualization.
- Multiple VMs can directly use the I/O device of the physical server.
- However, VMM needs to intercept I/O requests delivered by each VM in real time and emulates the request to a real I/O device, which causes severe performance loss to the server.

2. Para virtualization

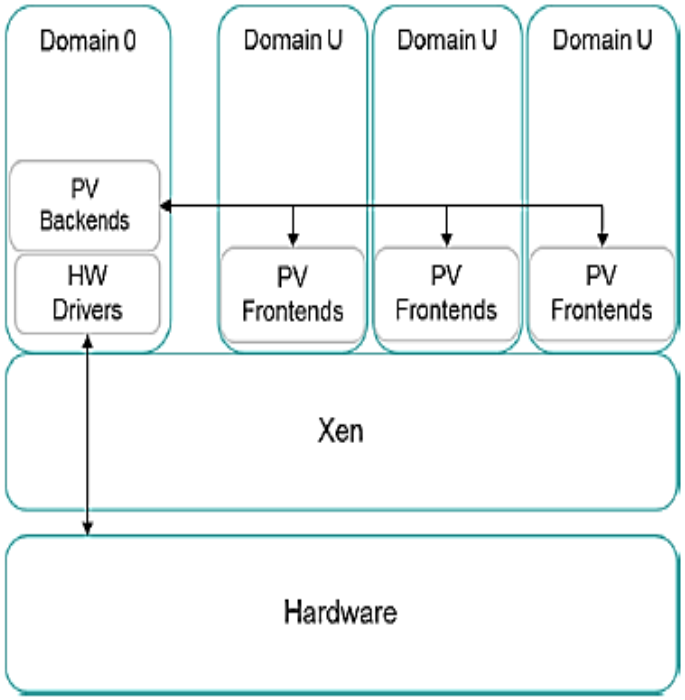
Unlike full virtualization, paravirtualization needs a **privileged VM.**

The Mechanism

1. Paravirtualization requires each VM to run a **frontend driver**.
2. When VMs need to access an I/O device, the VMs send I/O requests to the **privileged VM** through the frontend driver.
3. The backend driver of the privileged VM collects the I/O request sent by each VM.
4. Then, the backend driver processes multiple I/O requests by time and by channel.
5. The privileged VM runs the physical I/O device driver and sends the I/O request to the physical I/O device.
6. After processing the request, the I/O device returns the processing result to the privileged VM.

Para Virtualization Characteristics

- VMs send I/O requests to a privileged VM and then the privileged VM accesses a real I/O device. This **reduces the performance loss of VMM**.
- The VM OS needs to be modified. Specifically, the I/O request processing method of the OS needs to be changed so that all the I/O requests can be sent to the privileged VM for processing.
- This type can be used only with **open source OS's like Linux.**



3. Hardware-assisted virtualization

The Mechanism

Hardware-assisted **virtualization directly installs the I/O device driver in** the VM OS without any change to the OS.

Hardware-assisted Virtualization Characteristics

- This method is equivalent to traditional PC OS access to hardware. Therefore, the time required for a VM to access the I/O hardware is the same as that for a traditional PC to access the I/O hardware.

- Hardware-assisted virtualization **outperforms** full virtualization and paravirtualization in terms of I/O performance.

- However, hardware-assisted virtualization requires special hardware support.

Mainstream Compute Virtualization Technologies

In cloud computing, we mainly use virtualization to implement **IaaS** cloud services.

There are many mainstream virtualization technologies, generally:

Open-Source	Closed-source
Are free of charge and can be used anytime.	Are generally not free of charge and can be used out of the box.
Users can customize some special requirements based on open-source code	Users cannot view or customize source code.
Once a problem occurs in the system, the system recovery strongly relies on the administrator's skillset and experience.	If a system problem occurs, vendors provide all-round support.
Include KVM and Xen.	Include Microsoft Hyper-V, VMware vSphere, and Huawei FusionSphere.

Open Source Technology Software's	
KVM	**Xen**
KVM is full virtualization	Supports both paravirtualization and full virtualization.
It is **Type2 Hypervisor**, KVM, a module in the Linux kernel.	It is **Type1 Hypervisor** it is directly runs on hardware, and VMs run on Xen.
VMs Type on it: Common VM's	VMs Type on it: • Privileged VM • Common VM's
- It is used to virtualize CPUs and memory. - I/O devices (such as NICs and disks) need to be virtualized by QEMU.	- It is used to virtualize all compute resources.